MSG No. 190.232 $P_c\bar{6}2c$ [Type IV, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-6'2m'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,19,20,21,22,23,24]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[7,8,9,10,11,12,13,14,15,16,17,18]`

Table 2: Wyckoff site: `2b`, site symmetry: `-62'm'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,7,8,9,16,17,18,22,23,24]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[4,5,6,10,11,12,13,14,15,19,20,21]`

Table 3: Wyckoff site: `4c`, site symmetry: `-6'..`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,2,3,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[4,5,6,22,23,24]`
3	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[7,8,9,13,14,15]`
4	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[10,11,12,16,17,18]`

Table 4: Wyckoff site: `4d`, site symmetry: `-6..`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,2,3,7,8,9]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[4,5,6,10,11,12]`
3	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[13,14,15,19,20,21]`
4	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[16,17,18,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `3.m'`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,22,23,24]`
2	$[0, 0, -z]$	`[4,5,6,19,20,21]`
3	$\left[0, 0, \frac{1}{2} - z\right]$	`[7,8,9,16,17,18]`
4	$\left[0, 0, z + \frac{1}{2}\right]$	`[10,11,12,13,14,15]`

Table 6: Wyckoff site: `6f`, site symmetry: `m'2m'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,20,23]`
2	$[0, x, 0]$	`[2,5,21,24]`
3	$[-x, -x, 0]$	`[3,6,19,22]`

continued ...

Table 6

No.	position	mapping
4	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[7,10,15,18]
5	$\left[x,\ 0,\ \frac{1}{2}\right]$	[8,11,13,16]
6	$\left[0,\ x,\ \frac{1}{2}\right]$	[9,12,14,17]

Table 7: Wyckoff site: 6g, site symmetry: m2'm'

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,8,16,23]
2	$\left[0,\ x,\ \frac{1}{4}\right]$	[2,9,17,24]
3	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[3,7,18,22]
4	$\left[x,\ 0,\ \frac{3}{4}\right]$	[4,11,13,20]
5	$\left[0,\ x,\ \frac{3}{4}\right]$	[5,12,14,21]
6	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[6,10,15,19]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1,2,3]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[4,5,6]
3	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	[7,8,9]
4	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	[10,11,12]
5	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	[13,14,15]
6	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	[16,17,18]
7	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	[19,20,21]
8	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	[22,23,24]

Table 9: Wyckoff site: 12i, site symmetry: ..m'

No.	position	mapping
1	$[x,\ 0,\ z]$	[1,23]
2	$[0,\ x,\ z]$	[2,24]
3	$[-x,\ -x,\ z]$	[3,22]
4	$[x,\ 0,\ -z]$	[4,20]
5	$[0,\ x,\ -z]$	[5,21]
6	$[-x,\ -x,\ -z]$	[6,19]
7	$\left[-x,\ -x,\ \frac{1}{2}-z\right]$	[7,18]
8	$\left[x,\ 0,\ \frac{1}{2}-z\right]$	[8,16]
9	$\left[0,\ x,\ \frac{1}{2}-z\right]$	[9,17]
10	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	[10,15]
11	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	[11,13]
12	$\left[0,\ x,\ z+\frac{1}{2}\right]$	[12,14]

Table 10: Wyckoff site: `12j`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,20]`
2	$[-y,\ x-y,\ 0]$	`[2,21]`
3	$[-x+y,\ -x,\ 0]$	`[3,19]`
4	$[x-y,\ -y,\ 0]$	`[4,23]`
5	$[y,\ x,\ 0]$	`[5,24]`
6	$[-x,\ -x+y,\ 0]$	`[6,22]`
7	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[7,15]`
8	$\left[x,\ y,\ \frac{1}{2}\right]$	`[8,13]`
9	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[9,14]`
10	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[10,18]`
11	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[11,16]`
12	$\left[y,\ x,\ \frac{1}{2}\right]$	`[12,17]`

Table 11: Wyckoff site: `12k`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,8]`
2	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	`[2,9]`
3	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	`[3,7]`
4	$\left[x-y,\ -y,\ \frac{3}{4}\right]$	`[4,11]`
5	$\left[y,\ x,\ \frac{3}{4}\right]$	`[5,12]`
6	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	`[6,10]`
7	$\left[x,\ y,\ \frac{3}{4}\right]$	`[13,20]`
8	$\left[-y,\ x-y,\ \frac{3}{4}\right]$	`[14,21]`
9	$\left[-x+y,\ -x,\ \frac{3}{4}\right]$	`[15,19]`
10	$\left[x-y,\ -y,\ \frac{1}{4}\right]$	`[16,23]`
11	$\left[y,\ x,\ \frac{1}{4}\right]$	`[17,24]`
12	$\left[-x,\ -x+y,\ \frac{1}{4}\right]$	`[18,22]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`
3	$[-x+y,\ -x,\ z]$	`[3]`
4	$[x-y,\ -y,\ -z]$	`[4]`
5	$[y,\ x,\ -z]$	`[5]`
6	$[-x,\ -x+y,\ -z]$	`[6]`
7	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	`[7]`
8	$\left[x,\ y,\ \frac{1}{2}-z\right]$	`[8]`
9	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	`[9]`
10	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	`[11]`
12	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[12]`

continued ...

Table 12

No.	position	mapping
13	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[13]
14	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	[16]
17	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[17]
18	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	[18]
19	$[-x+y,\, -x,\, -z]$	[19]
20	$[x,\, y,\, -z]$	[20]
21	$[-y,\, x-y,\, -z]$	[21]
22	$[-x,\, -x+y,\, z]$	[22]
23	$[x-y,\, -y,\, z]$	[23]
24	$[y,\, x,\, z]$	[24]